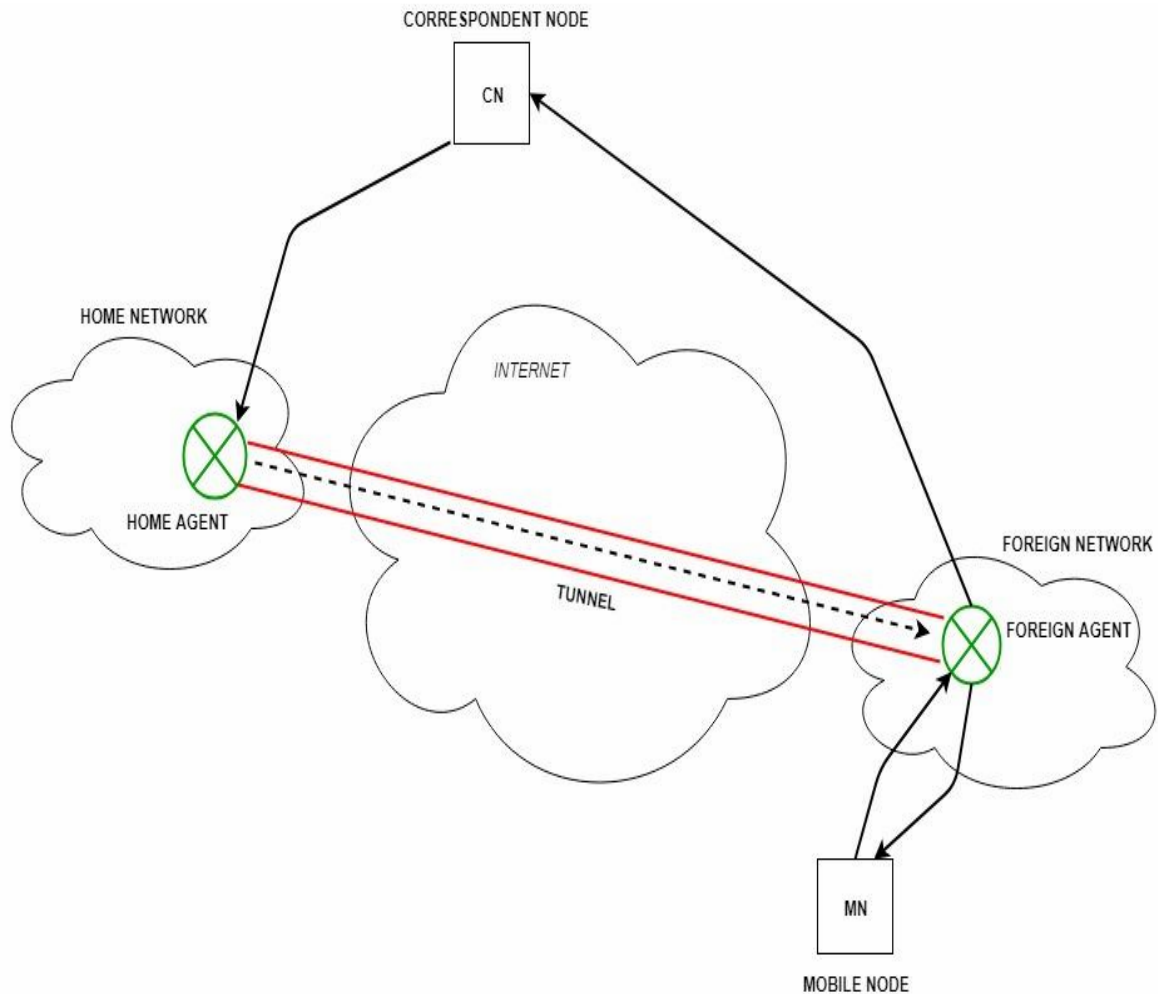


Mobile Internet Protocol (or Mobile IP)

Mobile IP is a communication protocol (created by extending Internet Protocol, IP) that allows the users to move from one network to another with the same IP address. It ensures that the communication will continue without the user's sessions or connections being dropped.

Terminologies:

1. **Mobile Node (MN)** is the hand-held communication device that the user carries e.g. Cell phone.
2. **Home Network** is a network to which the mobile node originally belongs as per its assigned IP address (home address).
3. **Home Agent (HA)** is a router in-home network to which the mobile node was originally connected
4. **Home Address** is the permanent IP address assigned to the mobile node (within its home network).
5. **Foreign Network** is the current network to which the mobile node is visiting (away from its home network).
6. **Foreign Agent (FA)** is a router in a foreign network to which the mobile node is currently connected. The packets from the home agent are sent to the foreign agent which delivers them to the mobile node.
7. **Correspondent Node (CN)** is a device on the internet communicating to the mobile node.
8. **Care-of Address (COA)** is the temporary address used by a mobile node while it is moving away from its home network.
9. **Foreign agent COA**, the COA could be located at the FA, i.e., the COA is an IP address of the FA. The FA is the tunnel end-point and forwards packets to the MN. Many MN using the FA can share this COA as a common COA.
10. **Co-located COA**, the COA is co-located if the MN temporarily acquired an additional IP address which acts as COA. This address is now topologically correct, and the tunnel endpoint is at the MN. Co-located addresses can be acquired using services such as DHCP.



Mobile IP

Working:

The correspondent node sends the data to the mobile node. Data packets contain the correspondent node's address (Source) and home address (Destination). Packets reach the home agent. But now mobile node is not in the home network, it has moved into the foreign network. The foreign agent sends the care-of-address to the home agent to which all the packets should be sent. Now, a tunnel will be established between the home agent and the foreign agent by the process of tunneling.

[Tunneling](#) establishes a virtual pipe for the packets available between a tunnel entry and an endpoint. It is the process of sending a packet via a tunnel and it is achieved by a mechanism called encapsulation.

Now, the home agent encapsulates the data packets into new packets in which the source address is the home address and destination is the care-of-address and sends it through the tunnel to the foreign agent. Foreign agent, on another side of the tunnel, receives the data packets, decapsulates them, and sends them to the mobile node. The mobile node in response to the data

packets received sends a reply in response to the foreign agent. The foreign agent directly sends the reply to the correspondent node.

Process of Mobile IP

The mobile IP process has following three main phases, which are:

1. Agent Discovery

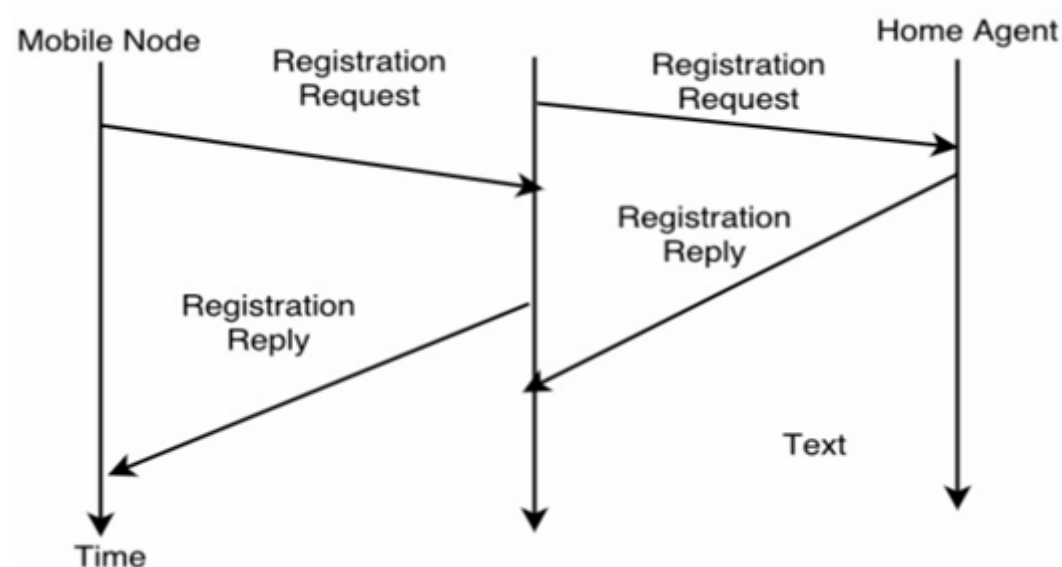
During the agent discovery phase the HA and FA advertise their services on the network by using the ICMP router discovery protocol (IROP).

Mobile IP defines two methods: agent advertisement and agent solicitation which are in fact router discovery methods plus extensions.

- **Agent advertisement:** For the first method, FA and HA advertise their presence periodically using special agent advertisement messages. These messages advertisement can be seen as a beacon broadcast into the subnet. For this advertisement internet control message protocol (ICMP) messages according to RFC 1256, are used with some mobility extensions.
- **Agent solicitation:** If no agent advertisements are present or the inter arrival time is too high, and an MN has not received a COA, the mobile node must send agent solicitations. These solicitations are again bases on RFC 1256 for router solicitations.

2. Registration

The main purpose of the registration is to inform the home agent of the current location for correct forwarding of packets.



Registration can be done in two ways depending on the location of the COA.

- **If the COA is at the FA**, the MN sends its registration request containing the COA to the FA which is forwarding the request to the HA. The HA now set up a **mobility binding** containing the mobile node's home IP address and the current COA.

Additionally, the mobility binding contains the lifetime of the registration which is negotiated during the registration process. Registration expires automatically after the lifetime and is deleted; so a mobile node should register before expiration. After setting up the mobility binding, the HA send a reply message back to the FA which forwards it to the MN.

- **If the COA is co-located**, registration can be very simpler. The mobile node may send the request directly to the HA and vice versa. This by the way is also the registration procedure for MNs returning to their home network.

IPv6 - Mobility

When a host is connected to a link or network, it acquires an IP address and all communication take place using that IP address on that link. As soon as, the same host changes its physical location, that is, moves into another area / subnet / network / link, its IP address changes accordingly, and all the communication taking place on the host using old IP address, goes down.

IPv6 mobility provides a mechanism for the host to roam around different links without losing any communication/connection and its IP address.

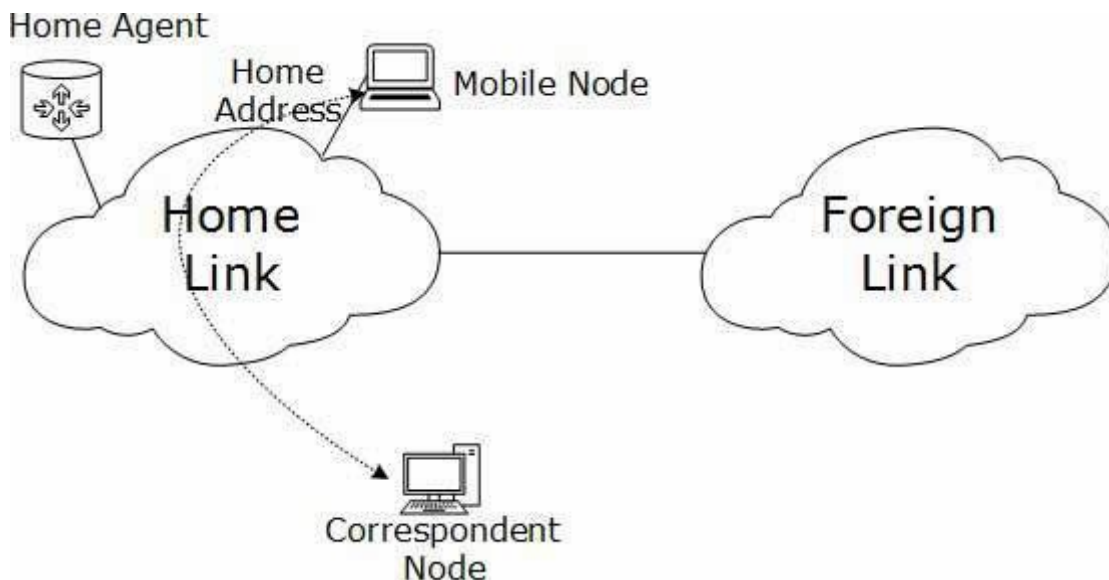
Multiple entities are involved in this technology:

- **Mobile Node:** The device that needs IPv6 mobility.
- **Home Link:** This link is configured with the home subnet prefix and this is where the Mobile IPv6 device gets its Home Address.
- **Home Address:** This is the address which the Mobile Node acquires from the Home Link. This is the permanent address of the Mobile Node. If the Mobile Node remains in the same Home Link, the communication among various entities take place as usual.
- **Home Agent:** This is a router that acts as a registrar for Mobile Nodes. Home Agent is connected to Home Link and maintains information about all Mobile Nodes, their Home Addresses, and their present IP addresses.

- **Foreign Link:** Any other Link that is not Mobile Node's Home Link.
- **Care-of Address:** When a Mobile Node gets attached to a Foreign Link, it acquires a new IP address of that Foreign Link's subnet. Home Agent maintains the information of both Home Address and Care-of Address. Multiple Care-of addresses can be assigned to a Mobile Node, but at any instance, only one Care-of Address has binding with the Home Address.
- **Correspondent Node:** Any IPv6 enabled device that intends to have communication with Mobile Node.

Mobility Operation

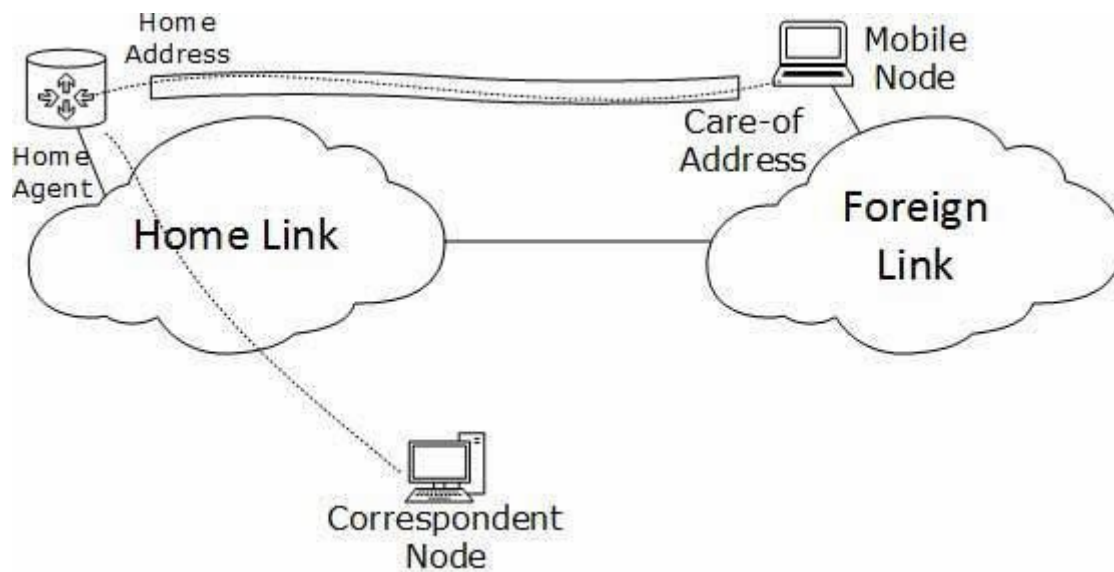
When Mobile Node stays in its Home Link, all communications take place on its Home Address as shown below:



[Image: Mobile Node connected to Home Link]

When a Mobile Node leaves its Home Link and is connected to some Foreign Link, the Mobility feature of IPv6 comes into play. After getting connected to a Foreign Link, the Mobile Node acquires an IPv6 address from the Foreign Link. This address is called Care-of Address. The Mobile Node sends a binding request to its Home Agent with the new Care-of Address. The Home Agent binds the Mobile Node's Home Address with the Care-of Address, establishing a Tunnel between both.

Whenever a Correspondent Node tries to establish connection with the Mobile Node (on its Home Address), the Home Agent intercepts the packet and forwards to Mobile Node's Care-of Address over the Tunnel which was already established.



[Image: Mobile Node connected to Foreign Link]

Route Optimization

When a Correspondent Node initiates a communication by sending packets to Mobile the Node on the Home Address, these packets are tunnelled to the Mobile Node by the Home Agent. In Route Optimization mode, when the Mobile Node receives a packet from the Correspondent Node, it does not forward replies to the Home Agent. Rather, it sends its packet directly to the Correspondent Node using Home Address as Source Address. This mode is optional and not used by default.